

DIABETES

DEFINITION

Diabetes is a chronic metabolic disorder in which the body either does not produce enough insulin or cannot use insulin effectively. This leads to high levels of glucose (sugar) in the blood.

TYPES

- Type 1 Diabetes
- Type 2 Diabetes
- Gestational Diabetes

SYMPTOMS

- Frequent urination
- Excessive thirst
- Increased hunger
- Unexplained weight loss
- Fatigue
- Blurred vision
- Slow healing wounds
- Frequent infections

CAUSES

TYPE 1 DIABETES

- Autoimmune destruction of insulin-producing cells

TYPE 2 DIABETES

- Obesity
- Physical inactivity
- Family history
- Insulin resistance

RISK FACTORS

- Overweight or obesity
- Family history of diabetes
- High blood pressure
- Poor diet
- Sedentary lifestyle

DIAGNOSIS

- Fasting Blood Sugar Test
- HbA1c Test
- Oral Glucose Tolerance Test
- Random Blood Sugar Test

TREATMENT

- Healthy diet
- Regular exercise
- Weight management
- Oral medications
- Insulin therapy if required

PREVENTION

- Maintain healthy weight
- Exercise regularly
- Avoid excessive sugar intake
- Eat balanced meals

COMPLICATIONS

- Heart disease
- Stroke
- Kidney disease
- Eye damage
- Nerve damage
- Foot ulcers

WHEN TO SEE A DOCTOR

- Persistent high blood sugar
- Excessive thirst and urination
- Sudden vision changes
- Non-healing wounds

MALARIA

DEFINITION

Malaria is a mosquito-borne infectious disease caused by Plasmodium parasites. It spreads through the bite of infected female Anopheles mosquitoes.

SYMPTOMS

- High fever
- Chills
- Sweating
- Headache
- Muscle pain
- Nausea
- Vomiting
- Fatigue

CAUSES

- Bite from infected mosquito
- Rarely through blood transfusion

RISK FACTORS

- Living in tropical regions
- Poor mosquito control
- Lack of protective measures

DIAGNOSIS

- Blood smear examination
- Rapid diagnostic tests (RDT)

TREATMENT

- Antimalarial medications
- Hydration
- Rest
- Hospitalization in severe cases

PREVENTION

- Use mosquito nets
- Apply mosquito repellents
- Wear full-sleeved clothing
- Eliminate stagnant water

COMPLICATIONS

- Severe anemia
- Organ failure
- Cerebral malaria
- Death if untreated

WHEN TO SEE A DOCTOR

- Fever after mosquito exposure
 - Persistent chills
 - Severe weakness
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DENGUE

DEFINITION

Dengue is a viral infection transmitted by Aedes mosquitoes. It is common in tropical and subtropical regions and usually occurs during rainy seasons when mosquito breeding increases. Dengue can range from a mild flu-like illness to a severe life-threatening condition known as Dengue Hemorrhagic Fever (DHF) or Dengue Shock Syndrome (DSS). Early diagnosis and proper hydration are important to prevent complications.

SYMPTOMS

- High fever
- Severe headache
- Pain behind the eyes
- Joint pain
- Muscle pain
- Skin rash
- Nausea
- Vomiting

CAUSES

- Dengue virus infection through mosquito bites

RISK FACTORS

- Living in dengue-prone regions
- Water accumulation around homes

DIAGNOSIS

- NS1 Antigen Test
- Dengue IgM and IgG Tests
- Complete Blood Count (CBC)

TREATMENT

- No specific antiviral treatment

- Rest
- Hydration
- Paracetamol for fever

PREVENTION

- Avoid mosquito bites
- Use mosquito repellents
- Keep surroundings clean
- Remove standing water

COMPLICATIONS

- Dengue Hemorrhagic Fever
- Dengue Shock Syndrome
- Internal bleeding
- Organ damage

EMERGENCY WARNING SIGNS

- Severe abdominal pain
- Continuous vomiting
- Difficulty breathing
- Bleeding gums
- Blood in stool or vomit

WHEN TO SEE A DOCTOR

- Persistent high fever
 - Severe weakness
 - Any warning signs mentioned above
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TYPHOID

DEFINITION

Typhoid fever is a bacterial infection caused by *Salmonella Typhi* and is usually spread through contaminated food and water. It spreads mainly through contaminated food and water and is more common in areas with poor sanitation and inadequate access to clean drinking water. The disease primarily affects the digestive system and can cause prolonged fever, weakness, and abdominal problems. If left untreated, typhoid can lead to severe complications such as intestinal bleeding and perforation.

SYMPTOMS

- Prolonged fever
- Weakness
- Headache
- Abdominal pain
- Loss of appetite
- Diarrhea or constipation

CAUSES

- Contaminated food
- Contaminated water
- Poor sanitation

RISK FACTORS

- Unsafe drinking water
- Poor hygiene
- Travel to endemic areas

DIAGNOSIS

- Blood culture
- Stool culture
- Widal test

TREATMENT

- Antibiotics prescribed by a doctor
- Adequate hydration
- Rest

PREVENTION

- Drink clean water
- Wash hands frequently
- Eat properly cooked food
- Maintain hygiene

COMPLICATIONS

- Intestinal bleeding
- Intestinal perforation
- Severe infection

WHEN TO SEE A DOCTOR

- Fever lasting more than 3 days

- Severe abdominal pain
 - Persistent weakness
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CHOLERA

DEFINITION

Cholera is an acute diarrheal disease caused by *Vibrio cholerae* bacteria. The disease mainly spreads through contaminated water and food and is often associated with poor sanitation and overcrowded living conditions. Cholera causes severe watery diarrhea that can rapidly lead to dehydration and electrolyte imbalance. Without prompt treatment, severe cholera can become life-threatening within a short period of time. However, with proper rehydration therapy, most patients recover successfully.

SYMPTOMS

- Severe watery diarrhea
- Vomiting
- Dehydration
- Muscle cramps
- Weakness

CAUSES

- Contaminated water
- Contaminated food

DIAGNOSIS

- Stool testing
- Clinical evaluation

TREATMENT

- Oral Rehydration Solution (ORS)
- Intravenous fluids
- Antibiotics when necessary

PREVENTION

- Drink safe water
- Maintain proper sanitation
- Wash hands regularly
- Safe food preparation

COMPLICATIONS

- Severe dehydration
- Kidney failure
- Shock
- Death if untreated

EMERGENCY WARNING SIGNS

- Extreme dehydration
- Confusion
- Rapid heartbeat
- Minimal urination
- COVID-19

DEFINITION

COVID-19 is a contagious respiratory illness caused by the SARS-CoV-2 virus. It can affect the lungs and other organs of the body. The virus spreads primarily through respiratory droplets and close contact with infected individuals. COVID-19 can affect people of all ages and may range from mild symptoms to severe illness involving the lungs, heart, brain, and other organs. Some individuals may remain asymptomatic, while others can develop serious complications requiring hospitalization. Vaccination and preventive measures have played a major role in reducing severe disease and

SYMPTOMS

- Fever
- Dry cough
- Fatigue
- Sore throat
- Loss of taste or smell
- Headache
- Body aches
- Shortness of breath

CAUSES

- Infection with SARS-CoV-2 virus
- Close contact with infected individuals

RISK FACTORS

- Older age
- Diabetes
- Heart disease
- Obesity
- Weak immune system

DIAGNOSIS

- RT-PCR Test
- Rapid Antigen Test
- Clinical Evaluation

TREATMENT

- Rest
- Hydration
- Symptom management
- Oxygen therapy in severe cases
- Hospital care when required

PREVENTION

- Hand hygiene
- Vaccination
- Avoid close contact with infected people
- Wear masks in high-risk situations

COMPLICATIONS

- Pneumonia
- Respiratory failure
- Long COVID
- Organ damage

EMERGENCY WARNING SIGNS

- Difficulty breathing
- Persistent chest pain
- Confusion
- Blue lips or face

WHEN TO SEE A DOCTOR

- Breathing problems
- High fever lasting several days

- Severe weakness
-

COMMON COLD

DEFINITION

The common cold is a mild viral infection affecting the nose and throat.

SYMPTOMS

- Runny nose
- Sneezing
- Sore throat
- Mild cough
- Nasal congestion
- Mild headache

CAUSES

- Rhinoviruses
- Other respiratory viruses

RISK FACTORS

- Close contact with infected people
- Weak immunity
- Seasonal changes

DIAGNOSIS

- Usually diagnosed clinically

TREATMENT

- Rest
- Fluids
- Steam inhalation
- Symptom-relieving medications

PREVENTION

- Frequent hand washing
- Avoid touching the face
- Avoid close contact with sick individuals

COMPLICATIONS

- Sinus infection
- Ear infection

- Worsening asthma symptoms

WHEN TO SEE A DOCTOR

- Symptoms lasting more than 10 days
 - High fever
 - Severe breathing problems
-

INFLUENZA (FLU)

DEFINITION

Influenza is a contagious viral infection that affects the respiratory system.

SYMPTOMS

- Sudden fever
- Chills
- Muscle aches
- Fatigue
- Headache
- Dry cough
- Sore throat

CAUSES

- Influenza virus infection

RISK FACTORS

- Elderly individuals
- Young children
- Pregnancy
- Chronic diseases

DIAGNOSIS

- Rapid Flu Test
- PCR Testing

TREATMENT

- Rest
- Hydration
- Antiviral medications when prescribed

PREVENTION

- Annual flu vaccination

- Hand hygiene
- Avoid close contact with infected individuals

COMPLICATIONS

- Pneumonia
- Sinus infection
- Hospitalization
- Death in severe cases

EMERGENCY WARNING SIGNS

- Difficulty breathing
- Chest pain
- Severe dehydration

WHEN TO SEE A DOCTOR

- High-risk patients with flu symptoms
 - Persistent fever
 - Breathing difficulties
-

PNEUMONIA

DEFINITION

Pneumonia is an infection that causes inflammation of the air sacs in one or both lungs.

SYMPTOMS

- Fever
- Chills
- Productive cough
- Chest pain
- Shortness of breath
- Fatigue

CAUSES

- Bacterial infection
- Viral infection
- Fungal infection

RISK FACTORS

- Smoking

- Old age
- Weak immune system
- Chronic diseases

DIAGNOSIS

- Chest X-ray
- Blood tests
- Sputum examination

TREATMENT

- Antibiotics for bacterial pneumonia
- Antiviral treatment if needed
- Oxygen therapy
- Hospitalization in severe cases

PREVENTION

- Vaccination
- Smoking cessation
- Good hygiene practices

COMPLICATIONS

- Respiratory failure
- Sepsis
- Lung abscess

EMERGENCY WARNING SIGNS

- Severe breathing difficulty
- Low oxygen levels
- Confusion

WHEN TO SEE A DOCTOR

- Persistent cough with fever
- Chest pain
- Difficulty breathing

TUBERCULOSIS (TB)

DEFINITION

Tuberculosis is a bacterial infection caused by *Mycobacterium tuberculosis*, primarily affecting the lungs.

SYMPTOMS

- Persistent cough lasting more than 2 weeks
- Fever
- Night sweats
- Weight loss
- Fatigue
- Coughing blood

CAUSES

- Airborne transmission from infected individuals

RISK FACTORS

- HIV infection
- Malnutrition
- Diabetes
- Smoking
- Crowded living conditions

DIAGNOSIS

- Sputum Test
- Chest X-ray
- TB Skin Test
- GeneXpert Test

TREATMENT

- Long-term antibiotic therapy
- Directly Observed Treatment (DOTS)

PREVENTION

- Early diagnosis
- BCG vaccination
- Good ventilation
- Covering mouth while coughing

COMPLICATIONS

- Lung damage
- Spread to other organs
- Drug-resistant TB

EMERGENCY WARNING SIGNS

- Coughing significant amounts of blood
- Severe breathing difficulty
- Extreme weakness

WHEN TO SEE A DOCTOR

- Persistent cough for more than 2 weeks
 - Unexplained weight loss
 - Night sweats
-

ASTHMA

DEFINITION

Asthma is a chronic disease that causes inflammation and narrowing of the airways.

SYMPTOMS

- Wheezing
- Shortness of breath
- Chest tightness
- Coughing, especially at night

CAUSES

- Genetic factors
- Allergies
- Environmental triggers

RISK FACTORS

- Family history
- Allergic conditions
- Air pollution
- Smoking exposure

DIAGNOSIS

- Spirometry
- Peak Flow Measurement
- Clinical Assessment

TREATMENT

- Inhalers
- Bronchodilators
- Corticosteroids

- Trigger avoidance

PREVENTION

- Avoid smoke and dust
- Follow prescribed treatment plans
- Monitor symptoms regularly

COMPLICATIONS

- Severe asthma attacks
- Respiratory failure in rare cases

EMERGENCY WARNING SIGNS

- Severe breathing difficulty
- Blue lips
- Inability to speak full sentences

WHEN TO SEE A DOCTOR

- Frequent asthma attacks
- Poor symptom control
- Need for rescue inhaler more often than usual

BRONCHITIS

DEFINITION

Bronchitis is inflammation of the bronchial tubes that carry air to and from the lungs.

SYMPTOMS

- Persistent cough
- Mucus production
- Wheezing
- Chest discomfort
- Fatigue

CAUSES

- Viral infections
- Smoking
- Air pollution

RISK FACTORS

- Smoking
- Exposure to dust and chemicals

- Weak immune system

DIAGNOSIS

- Physical examination
- Chest X-ray if needed

TREATMENT

- Rest
- Hydration
- Cough medications
- Bronchodilators when prescribed

PREVENTION

- Avoid smoking
- Hand hygiene
- Vaccination against flu

COMPLICATIONS

- Chronic bronchitis
- Pneumonia

WHEN TO SEE A DOCTOR

- Cough lasting more than 3 weeks
 - Blood in mucus
 - Breathing difficulty
-

HYPOTHYROIDISM

DEFINITION

Hypothyroidism is a condition in which the thyroid gland does not produce enough thyroid hormones, causing the body's metabolism to slow down.

SYMPTOMS

- Fatigue
- Weight gain
- Dry skin
- Hair loss
- Constipation
- Depression
- Cold intolerance

- Slow heart rate
- Memory problems

CAUSES

- Autoimmune thyroid disease (Hashimoto's Thyroiditis)
- Thyroid surgery
- Radiation therapy
- Iodine deficiency
- Certain medications

RISK FACTORS

- Female gender
- Age above 60 years
- Family history of thyroid disease
- Autoimmune disorders

DIAGNOSIS

- TSH Blood Test
- Free T4 Test
- Thyroid Antibody Tests

TREATMENT

- Thyroid hormone replacement therapy (Levothyroxine)
- Regular monitoring of thyroid levels

PREVENTION

- Adequate iodine intake
- Regular health checkups

COMPLICATIONS

- Heart disease
- Infertility
- Goiter
- Myxedema (rare but serious)

WHEN TO SEE A DOCTOR

- Persistent fatigue
 - Unexplained weight gain
 - Depression without clear cause
-

HYPERTHYROIDISM

DEFINITION

Hyperthyroidism occurs when the thyroid gland produces excessive amounts of thyroid hormones.

SYMPTOMS

- Weight loss
- Rapid heartbeat
- Anxiety
- Tremors
- Excessive sweating
- Heat intolerance
- Irritability
- Difficulty sleeping

CAUSES

- Graves' Disease
- Thyroid nodules
- Thyroiditis

RISK FACTORS

- Family history
- Female gender
- Autoimmune disorders

DIAGNOSIS

- TSH Test
- Free T3 and T4 Tests
- Thyroid Scan

TREATMENT

- Anti-thyroid medications
- Radioactive iodine therapy
- Surgery in selected cases

PREVENTION

- Regular thyroid monitoring in high-risk individuals

COMPLICATIONS

- Heart rhythm disorders

- Osteoporosis
- Thyroid Storm (medical emergency)

EMERGENCY WARNING SIGNS

- Very high fever
- Severe rapid heartbeat
- Confusion

WHEN TO SEE A DOCTOR

- Unexplained weight loss
 - Persistent rapid heartbeat
 - Severe anxiety symptoms
-

PCOS (POLYCYSTIC OVARY SYNDROME)

DEFINITION

PCOS is a hormonal disorder affecting women of reproductive age, often causing irregular menstrual cycles and excess androgen levels.

SYMPTOMS

- Irregular periods
- Excess facial hair
- Acne
- Weight gain
- Difficulty conceiving
- Hair thinning

CAUSES

- Hormonal imbalance
- Insulin resistance
- Genetic factors

RISK FACTORS

- Family history
- Obesity
- Insulin resistance

DIAGNOSIS

- Medical history
- Ultrasound examination

- Hormone testing

TREATMENT

- Weight management
- Hormonal medications
- Lifestyle modifications
- Fertility treatment if needed

PREVENTION

- Healthy diet
- Regular exercise
- Weight control

COMPLICATIONS

- Infertility
- Type 2 Diabetes
- Heart disease
- Sleep apnea

WHEN TO SEE A DOCTOR

- Irregular periods
- Difficulty becoming pregnant
- Excessive hair growth

GROWTH HORMONE DEFICIENCY

DEFINITION

Growth Hormone Deficiency occurs when the pituitary gland produces insufficient growth hormone.

SYMPTOMS

CHILDREN

- Short stature
- Delayed growth
- Delayed puberty

ADULTS

- Reduced muscle mass
- Increased body fat
- Low energy levels

CAUSES

- Pituitary gland disorders
- Brain injury
- Congenital conditions

DIAGNOSIS

- Growth hormone stimulation tests
- MRI scan
- Blood tests

TREATMENT

- Growth hormone replacement therapy

COMPLICATIONS

- Delayed development in children
- Reduced quality of life in adults

WHEN TO SEE A DOCTOR

- Poor growth in children
 - Unexplained fatigue in adults
-

GROWTH HORMONE EXCESS

DEFINITION

Growth Hormone Excess occurs when the body produces too much growth hormone.

SYMPTOMS

CHILDREN (GIGANTISM)

- Excessive height growth
- Enlarged hands and feet

ADULTS (ACROMEGALY)

- Enlarged hands
- Enlarged feet
- Enlarged jaw
- Thickened facial features

CAUSES

- Pituitary gland tumors

DIAGNOSIS

- Growth hormone testing
- MRI scan

TREATMENT

- Surgery
- Medications
- Radiation therapy

COMPLICATIONS

- Heart disease
- Diabetes
- Joint problems

WHEN TO SEE A DOCTOR

- Abnormal growth patterns
 - Enlarged facial features
-

ADDISON'S DISEASE

DEFINITION

Addison's Disease occurs when the adrenal glands do not produce enough cortisol and aldosterone.

SYMPTOMS

- Fatigue
- Weight loss
- Low blood pressure
- Muscle weakness
- Salt craving
- Darkening of skin

CAUSES

- Autoimmune disease
- Infections
- Adrenal gland damage

RISK FACTORS

- Autoimmune disorders

DIAGNOSIS

- Blood tests
- ACTH stimulation test

TREATMENT

- Hormone replacement therapy

PREVENTION

- No known prevention

COMPLICATIONS

- Addisonian Crisis (medical emergency)

EMERGENCY WARNING SIGNS

- Severe weakness
- Vomiting
- Confusion
- Loss of consciousness

WHEN TO SEE A DOCTOR

- Persistent fatigue
 - Unexplained weight loss
 - Low blood pressure
-

CUSHING SYNDROME

DEFINITION

Cushing Syndrome occurs when the body is exposed to excessive cortisol levels for a prolonged period.

SYMPTOMS

- Weight gain
- Moon-shaped face
- High blood pressure
- Easy bruising
- Muscle weakness
- Purple stretch marks

CAUSES

- Long-term steroid use
- Adrenal tumors
- Pituitary tumors

DIAGNOSIS

- Cortisol testing
- Imaging studies

TREATMENT

- Reduce steroid use if possible
- Surgery
- Medications

PREVENTION

- Use steroid medications only under medical supervision

COMPLICATIONS

- Diabetes
- High blood pressure
- Osteoporosis

WHEN TO SEE A DOCTOR

- Rapid unexplained weight gain
 - Severe muscle weakness
-

INSULIN RESISTANCE

DEFINITION

Insulin Resistance is a condition in which body cells do not respond properly to insulin, causing elevated blood sugar levels.

SYMPTOMS

- Often no symptoms initially
- Weight gain
- Fatigue
- Darkened skin patches (Acanthosis Nigricans)

CAUSES

- Obesity
- Physical inactivity
- Genetic factors

RISK FACTORS

- PCOS
- Family history of diabetes
- Overweight or obesity

DIAGNOSIS

- Fasting glucose
- HbA1c

- Insulin level testing

TREATMENT

- Weight loss
- Exercise
- Healthy diet
- Medications when prescribed

PREVENTION

- Active lifestyle
- Balanced diet
- Weight management

COMPLICATIONS

- Prediabetes
- Type 2 Diabetes
- Heart disease

WHEN TO SEE A DOCTOR

- Persistent weight gain
 - Elevated blood sugar levels
-

PREDIABETES

DEFINITION

Prediabetes is a condition where blood sugar levels are higher than normal but not high enough to be diagnosed as Type 2 Diabetes.

SYMPTOMS

- Often no symptoms
- Increased thirst
- Increased urination
- Fatigue

CAUSES

- Insulin resistance

RISK FACTORS

- Obesity
- Lack of exercise
- Family history of diabetes

DIAGNOSIS

- HbA1c Test
- Fasting Blood Sugar Test
- Glucose Tolerance Test

TREATMENT

- Weight loss
- Regular exercise
- Healthy eating habits

PREVENTION

- Healthy lifestyle
- Regular health screenings

COMPLICATIONS

- Type 2 Diabetes
- Heart disease

WHEN TO SEE A DOCTOR

- Elevated blood sugar readings
 - Diabetes risk factors
-